Your BOQ agent: what we fixed, and two questions for you

For Ajay | from Girish | 28 September 2026

Project tested: Proposed Residential Building (B+G+5+Roof), Plot 5310319, Saih Shuaib 2 - 102 drawings

The short version

Before: when you ran the agent on this project, it went through every step but **did not produce a Bill of Quantities**. It listed 162 problems.

Now: it **produces the Bill of Quantities** for the doors and windows: **480 doors and 199 windows** in 34 items, each with its size and type, laid out in the house format. The list of problems is down to a few real questions for the consultant.

What we need from you: two decisions, on pages 4 and 5. Each is a simple choice - just tick a box.

A few words used in this note

Word	What it means here
BOQ (Bill of Quantities)	The list of every item of work with how many or how much, for contractors to price.
Take-off	The working sheet behind the BOQ: where each number came from.
Schedule	A drawing that lists the doors (or windows) of a building: each type, its size and how many.
Title block	The box in the corner of every drawing with its number, name, scale and date.
Agent	The program you asked us to test, which reads the drawings and writes the BOQ.

What was going wrong

Four problems stopped the agent. None of them was a fault in your instructions. This consultant simply draws things differently from the drawings the agent was built on.

1. It could not read the drawing numbers

Every drawing has a title block. On this project the box says **"DRAWING NO."** with the number written underneath. The agent only knew the shorter "DWG NO.", so it missed the number on almost every drawing and fell back on the file names. It also cut long numbers short: **A-13-001** became just **A-13**, so three different drawings looked like the same one.

2. It did not know which drawings were structural

The structural drawings are simply called "STRUCTURE Sheet 05" and so on, with no letter to say what trade they belong to. The agent could not tell, so it could not look for anything on them.

3. It read the wrong table as the door schedule

On this project the door and window schedules are **drawn as pictures**: each door is a small drawing with its size written inside the picture, not as typed text. The only typed table on those sheets was the small revision box in the corner ("Date / Description / By"). The agent took that box for the schedule, found no doors in it, and so had nothing to put in the bill.

4. Some drawings it could not sort

The cover sheet, the list of drawings and the perspective views are never measured, but the agent did not know that and reported each of them as a problem.

> All four are now fixed, and they were tested on this project's 102 drawings.

What the agent does now

- Reads the **drawing number and name** of every drawing, including this consultant's layout, and keeps long numbers whole. **All 102 drawings** now have their number; 100 also have their name (the other two have none printed).
- Knows the **structural drawings** are structural because they sit in the "Structure" folder.
- Ignores the title block and revision box when looking for a schedule, so it can no longer mistake them for one.
- Recognises schedules that are **drawn as pictures** and copies the sizes and quantities printed on them (see Decision 1 on the next page).
- Knows which drawings are never measured (cover, list of drawings, perspectives, general notes) and leaves them out.
- Writes each door and window in the bill with its **kind, size and type**, the way your house bills are written.

How a line in the bill changed

Before	Now
Doors, type D-1, complete in accordance with the Specifications and Drawings	*Doors, hinged door, size 1200mm x 2200mm, type D-1, complete in accordance with the Specifications and Drawings*
Ditto, type D-2	*Ditto, hinged door, size 1100mm x 2200mm, type D-2*

The size must be stated for doors and windows (it is a rule in the measurement standard, POMI). Before, it was left out.

The result on this project

	Your run	Now
Bill of Quantities produced	No	**Yes**
Doors in the bill	0	**480**
Windows in the bill	0	**199**
Drawings with their number read	about half, many wrong	**all 102**
Problems listed (without AI)	193	**59** - most need the AI switched on (prices), the rest are real questions for the consultant (page 6)

We checked that the quantities in the bill match the working sheet behind it exactly, item by item.

Decision 1: copying numbers off drawings

PLEASE DECIDE

The background. When the agent was built, a firm rule was set: **the AI must never read a number off a drawing.** The idea was that every quantity should come from written information or from measurement, never from a computer "looking" at a picture and guessing.

The problem. On this project the door and window schedules are pictures. The sizes and quantities are printed inside the drawing, not typed. With the rule as it is, the agent cannot read them at all, so it cannot produce a bill.

What we have done (for your approval). We allowed one narrow exception:

- The AI may **copy** what is printed on a drawn schedule (door type, size, how many) and in a title block (drawing number and name), the way a person would copy it from the sheet.
- It may **never measure** anything, add anything up, or guess.
- Every number copied this way is **marked in the take-off** as "copied from the drawing", so anyone checking the bill can see where it came from.
- Each copy is **saved** and tied to that exact drawing. If the drawing is revised, it is copied again.
- The copied quantities are still **checked against the door and window tags counted on the plans**.

Your choice:

☐ **Yes - allow it.** The agent copies printed values from drawn schedules and title blocks, clearly marked. The bill is produced, as on this project.

☐ **No - keep the original rule.** Drawn schedules are reported as missing and no door or window bill is produced until the consultant sends the schedules as an Excel file or a PDF with typed text.

Good to know

Either way, our next step is to have the agent **measure** the doors and windows directly from the floor plans - counting them and measuring their width from the lines on the drawing. Then the schedule would only be used to describe each door and to double-check the count.

Decision 2: the project name on the bill

PLEASE DECIDE

What happens now. The agent names the bill after the **folder the drawings are kept in**. The folder for this test was called "Test Drawigs", so the top of every bill page reads **"Test Drawigs"** and the file sent out is called "Test Drawigs - BOQ - Rev 00".

What the drawings actually say. Every title block on this project states the real details:

> *Proposed Residential Building (B+G+5+Roof) | Plot 5310319 | Owner: G F S Real Estate L.L.C*

Your choice:

- ☐ **A - Keep it as it is.** Before every run, someone renames the drawings folder to the project's name. Nothing changes in the agent.
- ☐ **B - Take the name from the drawings (our recommendation).** The agent reads the project name, plot number and owner from the title blocks and puts them on the bill. If it cannot find them, it uses the folder name as it does now.

We recommend B because clients' folders are rarely named the way a bill should be headed, and the title blocks are the official source.

Questions to put to the consultant

These are not faults in the agent. They are things the drawings themselves leave unclear, and the agent reports them instead of guessing.

- **Two structural drawings carry the same number, S-02_4** (the raft foundation sections and the ramp sections). The ramp drawing is probably meant to be S-02_5.
- **The perspectives drawing** is filed as A-02-001, but its title block says G-102. Which is right?
- **Door D-1:** the schedule says 20, but 19 are tagged on the plans.
- **Window DW 01:** the schedule says 1, but 2 are tagged on the plans.
- **Four items are for the basement**, but there is no basement floor plan in the set, so they cannot be checked.
- **Eight windows are L-shaped or in two parts** (DW 04, 06, 07, 09, 10, 11, 15 and 17). Their sizes are written in the bill exactly as printed, for example "(7300 + 2950)mm x 3200mm". Please confirm them.
- **The cover sheet** and one page of the area calculation have no drawing number printed.

What happens next

- Once you have ticked your two choices, send this back to Girish and we will set the agent accordingly.
- The updated agent is saved in the shared project on GitHub, with the team's latest improvements included.
- Our next piece of work is measuring the doors and windows straight from the floor plans (see page 4).

> If anything in this note is unclear, ask Girish - no technical knowledge is needed to make either decision.